Project Workflow

Overview

The Smart Lender project follows a structured machine learning workflow consisting of five major epics. Each epic represents a specific stage of the project lifecycle, beginning with data collection and ending with deployment of the loan prediction application. This systematic workflow ensures accurate model development, efficient implementation, and successful deployment.

Project Workflow

Start

│

▼

Epic 1: Data Collection and Architecture Design

│

▼

Epic 2: Visualizing and Analyzing the Data

│

▼

Epic 3: Data Pre-Processing

│

▼

Epic 4: Model Building

│

▼

Epic 5: Application Building

│

▼

End

Epic 1: Data Collection and Architecture Design

Objective

Collect the required dataset and design the overall architecture of the Smart Lender application.

Story 1: Dataset Collection

- Download the loan prediction dataset.
- Store the dataset in the project directory.
- Verify that the dataset is complete and accessible.

Story 2: Application Architecture Design

Define the application architecture, including:

- Data Flow
- Machine Learning Workflow
- Flask Application Structure
- Deployment Architecture

Deliverables

- Loan dataset
- Project folder structure
- Application architecture
- Workflow documentation

Epic 2: Visualizing and Analyzing the Data

Objective

Analyze the dataset to understand its characteristics and identify meaningful patterns.

Story 1: Import Dataset

- Import Pandas library.
- Read the CSV dataset.
- Display dataset information.

Story 2: Univariate Analysis

Analyze individual features using:

- Histograms
- Count plots
- Pie charts
- Box plots

Story 3: Bivariate Analysis

Analyze relationships between two variables using:

- Scatter plots
- Bar charts
- Correlation analysis

Story 4: Multivariate Analysis

Study relationships among multiple variables using:

- Correlation matrix
- Heatmap
- Pair plots

Deliverables

- Data analysis report
- Visualization graphs
- Feature insights

Epic 3: Data Pre-Processing

Objective

Prepare the dataset for machine learning model development.

Story 1: Data Cleaning

- Handle missing values.
- Remove duplicate records.
- Correct inconsistent data.

Story 2: Dataset Balancing

- Analyze class distribution.
- Balance target classes if required.

Story 3: Feature Scaling

- Normalize numerical features.
- Standardize data for better model performance.

Story 4: Train-Test Split

- Split dataset into training and testing sets.
- Prepare data for model training.

Deliverables

- Clean dataset
- Balanced dataset
- Scaled features
- Training and testing datasets

Epic 4: Model Building

Objective

Develop and compare multiple machine learning models to identify the best-performing model.

Story 1: Decision Tree

- Train Decision Tree classifier.
- Evaluate model performance.

Story 2: Random Forest

- Train Random Forest classifier.
- Compare with Decision Tree.

Story 3: K-Nearest Neighbors (KNN)

- Train KNN model.
- Analyze prediction accuracy.

Story 4: XGBoost

- Train XGBoost classifier.
- Compare with previous models.

Story 5: Model Evaluation

Evaluate all models using:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Save the best-performing model for deployment.

Deliverables

- Trained machine learning models
- Performance comparison
- Best model file
- Evaluation report

Epic 5: Application Building

Objective

Develop and deploy the Smart Lender web application.

Story 1: HTML Interface

Design user-friendly web pages, including:

- Home Page
- Prediction Page
- Result Page

Story 2: Flask Integration

- Build the Flask application.
- Connect the trained model.
- Handle user input.
- Display prediction results.

Story 3: Testing and Validation

- Run the application.
- Validate prediction accuracy.
- Test user interface.
- Verify application functionality.

Deliverables

- HTML templates
- Flask application
- Integrated ML model
- Working loan prediction system

Overall Workflow Diagram

Loan Dataset

│

▼

Data Collection

│

▼

Data Analysis

│

▼

Data Preprocessing

│

▼

Feature Engineering

│

▼

Model Training

│

▼

Model Evaluation

│

▼

Best Model Selection

│

▼

Flask Integration

│

▼

HTML User Interface

|
▼
Loan Prediction

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- XGBoost
- Flask
- HTML
- CSS
- JavaScript

Expected Outcome

Upon completion of the workflow:

- The dataset is collected and prepared.
- Data is analyzed and preprocessed.
- Multiple machine learning models are trained and evaluated.
- The best-performing model is selected.
- A Flask-based web application is developed.
- Users can enter loan application details and receive accurate loan approval predictions through an easy-to-use interface.

Conclusion

The Smart Lender project follows a structured workflow that covers every stage of the machine learning lifecycle, from data collection and preprocessing to model development and deployment. Organizing the project into clearly defined epics and stories ensures systematic development, improves maintainability, and supports the creation of an efficient loan prediction application.